

BARUUNTURUUN, Mongolia

WMO: 442130

Lat: 49.6586N

Lon: 94.4042E

Elev: 1232

StdP: 87.37

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.8	-38.8	-44.1	0.1	-40.7	-41.8	0.1	-38.6	4.7	-23.7	4.1	-26.7	0.0	120	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	30.2	16.7	28.2	15.8	26.4	15.2	19.0	26.7	17.6	25.5	16.5	24.0	3.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	13.3	23.6	14.6	12.1	21.9	13.3	11.1	20.5	59.8	26.9	54.9	25.3	50.9	24.1	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.5	7.0	5.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-40.8	34.0	2.8	2.8	-42.8	36.0	-44.4	37.6	-46.0	39.1	-48.0	41.1		
			-40.6	20.1	3.0	2.5	-42.7	21.9	-44.5	23.4	-46.1	24.8	-48.3	26.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.4	-29.9	-26.8	-15.5	2.2	10.7	17.0	19.2	16.9	10.2	1.4	-11.6	-24.1	(6)
(7)		DBStd	18.16	4.06	5.54	6.39	6.70	5.08	4.08	3.41	3.82	4.76	5.35	5.78	5.09	(7)
(8)		HDD10.0	5388	1237	1029	792	242	53	3	0	2	55	270	648	1056	(8)
(9)		HDD18.3	7692	1496	1263	1050	484	241	71	29	73	248	526	898	1314	(9)
(10)		CDD10.0	858	0	0	0	8	75	213	285	214	60	2	0	0	(10)
(11)		CDD18.3	119	0	0	0	0	3	31	56	27	2	0	0	0	(11)
(12)		CDH23.3	1326	0	0	0	0	54	369	577	302	24	0	0	0	(12)
(13)		CDH26.7	361	0	0	0	0	7	92	178	82	3	0	0	0	(13)
(14)	Wind	WSAvg	1.5	0.3	0.4	0.9	2.2	2.9	2.4	2.0	1.8	1.9	1.6	0.9	0.5	(14)
(15)	Precipitation	PrecAvg	219	4	4	7	15	19	30	50	40	23	13	10	7	(15)
(16)		PrecMax	312	12	12	17	35	54	85	146	118	63	37	30	16	(16)
(17)		PrecMin	109	0	0	0	2	1	2	5	5	4	0	1	0	(17)
(18)		PrecStd	42	3	3	4	9	13	17	32	22	14	9	7	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-17.4	-8.6	5.0	20.6	27.6	31.6	33.7	32.1	25.8	17.3	5.2	-7.9	(19)
(20)			MCWB	-18.1	-9.8	0.8	9.3	13.7	16.2	19.9	17.5	13.6	7.9	1.5	-9.3	(20)
(21)		2%	DB	-20.0	-12.8	-0.2	17.8	24.7	29.2	30.7	28.8	23.3	14.7	1.6	-12.1	(21)
(22)			MCWB	-20.4	-13.6	-2.3	8.7	12.7	15.3	17.7	16.2	12.7	6.8	-1.2	-12.9	(22)
(23)		5%	DB	-21.8	-16.3	-3.2	15.0	22.4	27.2	28.6	26.6	21.1	12.3	-0.8	-14.6	(23)
(24)			MCWB	-22.2	-16.9	-4.6	7.2	11.6	15.0	16.5	15.5	11.6	5.5	-2.5	-15.1	(24)
(25)		10%	DB	-23.5	-18.7	-5.9	12.5	20.1	25.4	26.6	24.5	18.7	9.9	-3.2	-16.6	(25)
(26)			MCWB	-23.8	-19.2	-7.0	5.9	10.5	14.4	15.7	14.9	10.4	4.2	-4.5	-17.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-18.0	-9.8	1.2	11.2	15.9	18.7	21.6	19.8	15.6	9.1	1.8	-9.1	(27)
(28)			MCDWB	-17.5	-8.7	5.3	18.5	24.5	26.9	31.5	29.7	24.7	15.7	4.8	-8.3	(28)
(29)		2%	WB	-20.4	-13.6	-2.2	9.0	14.1	17.4	19.8	17.9	13.6	7.4	-0.9	-12.8	(29)
(30)			MCDWB	-20.0	-12.8	-0.3	16.8	22.5	26.3	28.1	26.3	21.5	13.6	1.0	-12.2	(30)
(31)		5%	WB	-22.1	-16.8	-4.6	7.5	12.5	16.3	18.5	16.6	12.1	5.9	-2.5	-15.1	(31)
(32)			MCDWB	-21.8	-16.3	-3.3	14.5	20.9	25.0	25.7	24.5	19.8	11.9	-0.8	-14.7	(32)
(33)		10%	WB	-23.8	-19.2	-6.9	6.0	11.0	15.1	17.1	15.5	10.9	4.5	-4.6	-17.1	(33)
(34)			MCDWB	-23.6	-18.7	-5.9	12.1	19.2	23.5	24.0	22.6	17.7	9.6	-3.2	-16.7	(34)
(35)	Mean Daily Temperature Range	MDBR	9.3	10.6	11.5	10.4	12.0	12.0	11.0	11.0	11.7	10.8	9.1	8.5	(35)	
(36)		5% DB	MCDBR	10.0	12.0	11.3	13.3	14.4	14.6	14.4	14.1	13.9	13.7	9.7	9.5	(36)
(37)			MCWBR	9.7	11.2	9.4	7.3	7.2	6.6	6.4	6.6	7.0	8.4	7.5	9.0	(37)
(38)		5% WB	MCDBR	10.0	12.0	11.0	12.7	13.7	12.9	12.5	12.5	13.1	13.2	9.3	9.4	(38)
(39)			MCWBR	9.6	11.3	9.2	7.8	8.0	7.2	6.9	6.9	7.2	8.4	7.5	8.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.242	0.268	0.312	0.353	0.358	0.358	0.358	0.339	0.304	0.288	0.267	0.244	(40)	
(41)		taud	2.284	2.230	2.208	2.248	2.268	2.359	2.389	2.415	2.462	2.428	2.325	2.256	(41)	
(42)		Ebn at Noon	824	879	892	890	898	899	894	897	900	852	779	759	(42)	
(43)		Edh at Noon	68	93	115	125	128	118	113	105	89	76	65	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.42	2.46	4.05	5.44	6.56	6.83	6.36	5.53	4.45	2.80	1.48	1.08	(44)	
(45)		RadStd	0.11	0.13	0.13	0.27	0.31	0.25	0.35	0.28	0.24	0.17	0.12	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	-2.17	-3.33	N/A	-1.16	-1.37	N/A	N/A	N/A
(47)	Regional (5 neighbors)		N/A	-1.95	-2.68	N/A	-0.86	-0.90	N/A	N/A	N/A

Nomenclature: See separate page